

Assignment-1

Part 2

Software Testing

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Section: 4CS-S2

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Introduction

this report shows unit tests both that pass and fail for the 3 core classes , DigitScrubber , EmailScrubber and MainScrubber .

Test Classes

Test Class 1 (DigitScrubberTest)

Test 1: testScrubDigits_NormalCase

It tests the normal happy path of scrubbing numbers within a text

Result: passed 

Test 2: testScrubDigits_WithPrice_ShouldNotChange

it tests prices scrubbing which should not scrub

Result : fail 

Defect

Test 3: testScrubDigits_MixedInput

it tests both price and numbers together

Result : fail 

Defect

Test 4: testScrubDigits_NullInput_ShouldThrow

it tests that null pointer exception when string is null

Result: passed 

Test 5: testScrubDigits_BlankInput_ShouldThrow

it tests that illegal argument exception if string is blank

Result : passed 

Test Class 2 (EmailScrubberTest)

Test 1: testEmailScrub_NormalCase

test normal happy path , no exception and email doesn't have numbers

Result: passed 

Test 2: testScrubEmails_NoEmails

test if no emails, just normal text

Result: passed 

Test 3: testScrubEmails_MultipleEmails(

Test if more than one email

Result: passed 

Test 4: testScrubEmails_NullInput_ShouldThrow

Test throwing null pointer exception if text is null

Result: passed 

Test 5: testScrubEmails_BlankInput_ShouldThrow

Test throwing illegal argument exception if text is blank

Result : fail ❌

Defect

Test 6: testEmailScrub_EmailWithNumbers

Test email with a numbers in it , should scrub it

Result : fail ❌

Defect

Test Class 3 (MainScrubberTest)

Test 1: testDigitsMode

Test using only digits mode with a mock

Result: passed ✅

Test 2: testEmailsMode

Test using only email mode with a mock

Result: passed ✅

Test 3: testFullScrubbingMode

Test using both email mode and digit mode (full scrubbing) with a mock

Result: passed ✅

Test 4: testNullInput_ShouldThrow

Test if scrub input is null , it should throw null pointer exception

Result : fail ❌

Defect

Test 5: testBlankInput_ShouldThrow

Test if scrub input is blanks, it should throw illegal argument exception

Result : fail ❌

Defect

Defects

Defect 1: Price Handling in DigitScrubber

Test: testPriceNotScrubbed and testScrubDigits_MixedInput

Expected: "1111\$" remains unchanged

Actual: "XXX1\$"

Cause: Regex incorrectly replaces digits even when followed by \$

Defect 2: Exception Handling in EmailScrubber

Test: testBlankInput_ShouldThrowIllegalArgumentException

Expected: IllegalArgumentException

Actual: NullPointerException

Cause: Implementation treats blank input as null ,both throw null pointer exception

Defect 3: Email regex doesn't allow numbers

Test: testEmailScrub_EmailWithNumbers

Expected: "contact me at [EMAIL_HIDDEN]"

Actual: " contact me at Shazawagdy2004@gmail.com "

Cause: Email regex incorrectly takes numbers in emails from 0-0 , it should be from 0-9

Defect 4: Exception Suppression in MainScrubber

Test: testNullInput_ShouldThrow and testBlankInput_ShouldThrow

Expected: NullPointerException and IllegalArgumentException

Actual: No exception, method returns null

Cause: try-catch block suppresses exceptions and just return null in main scrubber class